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Design of a single-phase immersion cooling system through experimental and numerical analysis

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ABSTRACT

Cooling electronics systems more effectively could enable computer systems to utilize power more efficiently. Immersion cooling is a potential method to cool computers and other electronics by submerging them into a thermal conductive dielectric liquid or coolant. In this study, an innovative cooling structure and procedure of a single-phase immersion cooling system using heat sink and forced circulation is presented. The straight finned heat sink is attached on the CPU surface, and the entire mainboard is submerged in an engineered fluid, 3 M Novec 7100, which is able to dissipate heat and is used in immersion cooling applications. An experimental test is utilized to verify a simulation model built using the computer simulation software. Three circulating speeds of the liquid coolant along with the use of two different materials for the heat sink are chosen to explore the cooling effects of a single-phase immersion cooling system. The heat distribution of the designed model at various flow rates of liquid coolant and materials was observed through the cross-sectional viewpoint and 3D isothermal surface model. The results of the simulations show that the faster flowing speed of liquid coolant would remove more heat, and cause lower temperature of CPU. However, the flow of coolant was encumbered due to slower circulating speed. It also caused the higher temperature values and unbalanced heat distribution. The highest temperatures were measured, and the unbalanced heat distribution of the model was observed. It is noted that the material of the heat sink do not significantly affect the results. These outcomes are able to provide the system designer with useful information to increase power densification and guarantee the safe operation of the related computer, server and communication systems.

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1. Introduction

Cooling electronics systems efficiently could drive the power utilization more concentrated. The International Electronics Manufacturing Initiative (iNEMI) Technology Roadmap reports that how to draw the heat flux generated by chips within an area of potentially only 300 W/cm^2 while keeping the operating temperatures below 85 °C will be the main challenge for the micro and power-electronics industry [1]. Therefore, the cooling technology to dissipate the generated heat by chips and maintain the safe operation of electronic devices is a critical issue. Several methods, including thermoelectric cooler (TEC), heat pipe, vapor chamber, nanofluids, single-phase cooling, two-phase cooling, immersion cooling and vapor compression refrigeration (VCR) system are options for this purpose. The TEC coupling with thermosiphone loop [2] or heat sink [3] has higher efficiency and temperature uniformity comparing with other methods, and its performance increase with thermoelectric operating voltage and the dimension closeness between thermoelectric module and central processing unit (CPU). Heat pipes are able to dissipate substantial amount of heat with a relative small temperature drop along the heat pipe while providing a self-pumping ability due to an embedded porous material in their structure. The limiting factor for the heat transfer capability of a heat pipe is related to the working fluid transport capability [4]. Based on phase change heat transfer phenomena, vapor chamber draws high heat flux from a small heating source, and spreads to larger area through vaporization and condensation of working fluid inside it. The vapor chamber can keep the system at constant temperature nearly to avoid overheating or other damage. The thermal performance of vapor chamber is related to the

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Nomenclatures

Symbols

g	weight
Cp	specific heat capacity (J/kg•K)
J	joule
k	heat transfer coefficient (W/m^2K)
K	Kelvin temperature
m	meter
P	pressure (Pa)
q	heat flux (W/m^2)
Q	heating source
T	temperature (K)
V	average velocity (m/s)
W	watt (J/s)

Greek letters

μ	dynamic viscosity (Pa•s)
ρ	density (kg/m^3)
∇	cross product

Acronyms

3D	three dimensions
Al	aluminum
BDF	back difference in phase
CFD	computational fluid dynamics
CPU	central processing unit
Cu	copper
LHPs	loop heat pipes
PC	personal computer
TDP	thermal design power
TEC	thermoelectric cooler
VCR	vapor compression refrigeration

property of working fluid, material and geometric structure of wick structure [5]. The vapor chamber embedded with heat sinks with various fin configurations draw much attention in thermal management of electronics devices recently [6]. Nanofluids, the mixture of base fluid with nanoparticles, are able to improve the thermal conductivity and convection coefficients of fluid [7]. It is one of the candidates for further miniaturization of electronic devices. But a balance among volumetric concentration of nanoparticles, the flow rate and other variables to satisfy the economy and power consumption of cooling the system is still unconcluded [8]. Two-phase system of the micro-evaporator cooling cycles, composed by the pump [9], the compressor, the loop heat pipes (LHPs) [10] or the hybrid [11], could not only improve the heat removal efficiency but also permit the reuse of waste heat. It is suitable for a computer blade server, green data center and supercomputers, etc. VCR system, consisting compressor and heat exchanger (evaporator and condenser), can maintain low junction temperature while dissipate high heat flux [12]. Miniature VCR system can fit the requirement of limited space of today's electronic devices by combining the miniature compressor and microchannel heat exchanger. The main concern for VCR system is the cost [13].

Among them, immersion cooling is a potential method to cool computer and other electronics, such as complete servers and mainboard, by submerging them into a thermal conductive dielectric liquid or coolant [14]. This method removes the generated heat from the system by circulating the liquid throughout the system in order to continually dissipate heat that is generated by the components. The immersion cooling market is projected to increase from USD 177 million in 2019 to USD 501 million by 2024 [15] due to increased advances and uses in immersion edge computing, artificial intelligence, and cloud computing [16]. The requirement for the working fluid of an immersion cooling system is that it must be non-conductive with the electronics. Several types of fluids are suitable for cooling, however, the use of liquids is one of the most prominent and efficient candidates [14]. Four types of fluids, including deionized water, mineral oil, fluorocarbon-based fluids and synthetic oil, are suitable for this application. Depending on the properties of the fluid, the fluids used in the immersion cooling technology can be classified into a single-phase or two-phase systems. The difference between these two systems is whether the fluid that is used changes its state or not during the process of cooling the electronics. Matsuoka, et al. [17] evaluated the cooling performance of a liquid immersion cooling system with natural convection for high power server boards used in data center by computational fluid dynamics (CFD) software packages simulation and actual experiments. They tested several fluids, including silicone oil, soybean oil, and perfluorocarbon structured liquids, and found that the smoother refrigerant was better for cooling the high power CPU. Wagner, et al. [18] compared the cooling performance of three different liquid cooling technologies, including single-phase cold plate, two-phase immersion cooling with 3 M Novec® 649 and single-phase mineral oil immersion. The experimental results indicated that the single-phase cold plate achieve the highest power dissipation with the lowest thermal resistance of 0.048 K/W, the two-phase immersion cooling performed second best, then was the single-phase mineral oil immersion cooling. Sathyanarayana, et al. [19] presented the mixture formulations of Novec fluid (pure HFE 7200) with alcohols and ethers (HFE 7200 and methanol; HFE 7200 and ethoxybutane), which equipped low thermal conductivity and specific heat, to improve heat transfer properties and applicability of heat transfer fluids. The thermal performance of these new fluid mixtures was tested on a 1 cm × 1 cm silicon (Si) substrate having copper nanowire arrays at saturation conditions. Based on previous research results, fluorocarbon-based fluids evaporated in the largest amounts during the cooling process, and the mineral oil removed the least amount of heat from a system compared with other fluids, whereas the synthetic oil showed the best cooling performance.

Chen, et al. [20] installed radiation fins to the surface of a CPU to see if this could enhance heat-removing capabilities. They found that the novel three-dimensional (3D) stacked package structure with horizontal fins did in fact lead to better heat radiation capacity than the conventional design. Qiu, et al. [21] investigated the thermal behavior of 3D stacked dies using the cooling technologies of natural convection, forced air cooling and water immersion cooling methods. These models were built by using one of ANSYS analysis modules for electronic system thermal management, which is ICEPAK software. Simulation models were built for experimental validation and further thermal analysis. The results indicated that the thermal resistances of these three methods were 26, 7.6 and 0.6 °C/W, respectively. The immersion cooling method had the best cooling capability and the lowest temperature at the hotspot, which was the area of highest current density. An, et al. [22] compared a two-phase immersion cooling solution for high-powered processor designs with various heating sources by using ANSYS Fluent for a 3D numerical analysis. Two arrangements, including two vertically mounted heat sources to achieve higher packing density of the server and a single heat source, were immersed in a bath of 3 M Novec7000 as the phase change coolant. They found that Novec7000 was able to cool a 5 cm × 5 cm heat source in a vertical orientation with power as high as 225 W (heat flux 9 W/cm^2). In addition to ANSYS Fluent, the generally utilized CFD software packages for geophysical modeling also include COMSOL Multiphysics. Wang, et al. [23] investigated the electrical field distribution rule for direct current method forward modeling and point electric source by using COMSOL Multiphysics. Through comparing and analyzing COMSOL Multiphysics modeling results and

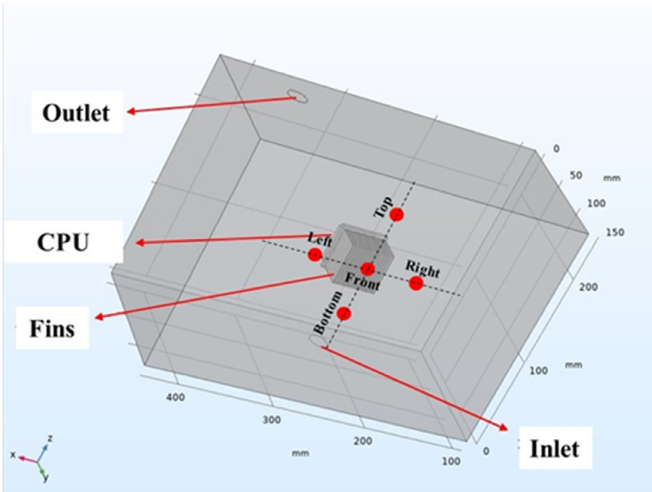

Fig. 1. Model structure of an immersion cooling system for the CPU of a personal computer.

theoretic values of typical models, they indicated that the validity and feasibility of direct current method forward modeling is proved and the COMSOL Multiphysics was suitable in geophysical modeling. By using COMSOL Multiphysics software, Kocman, et al. [24] also carried out a simulation of three-phase squirrel cage induction motor. They indicated that the simulation results from COMSOL match the catalog values rather well in the values of stator current and motor torque.

Based on the previous researches, an innovative cooling structure and procedure of a single-phase immersion cooling system with heat sink and forced circulation of liquid are presented in this study. The straight finned heat sink is attached on the CPU surface. The entire mainboard is immersed in the engineered 3 M Novec 7100 fluid for forced circulation cooling. An experimental test is utilized to verify the simulation model built by using computer simulation software. Three circulation speeds of the liquid coolant and two materials for the heat sink are chosen to explore the cooling effects of the designed single-phase immersion cooling system.

2. Model

2.1. Model configuration

In order to simulate the effect of the immersion cooling system with circulating liquid and heat sink placed on the CPU surface of a personal computer (PC), a three dimensional model was built using the computer simulation software, as shown in Fig. 1. The model includes a CPU, which is the main heat source of a PC, with a straight finned heat sink on its surface. The size of CPU is 40 mm-in-square. The chosen materials of the heat sink were aluminum (Al) and copper (Cu). The liquid coolant flowed into the PC box from the bottom of heat sink, passed the heat sink and flowed out through the top left of the box. The model was assumed as a steady, incompressible and laminar flow. The momentum equation was expressed as [25,26]:

$$\rho(\mathrm{DV/Dt}) = -\nabla P + \rho \mathrm{g} + \mu \nabla^2 \mathrm{V} \quad (1)$$

where ρ(DV/Dt) is the force, ∇P is the pressure, ρg is the weighting force, and $\mu\nabla^2$V is the viscosity force. All the forces are related to unit volume. The heat transfer function is expressed as [27–29]:

$$Q = \rho \mathrm{C_p}(\partial \mathrm{T}/\partial \mathrm{t}) + \rho \mathrm{C_p} \mathrm{u} \cdot \nabla T + \nabla \cdot \mathrm{q} \quad (2)$$

$$\vec{\mathrm{q}} = -k\nabla \mathrm{T} \quad (3)$$

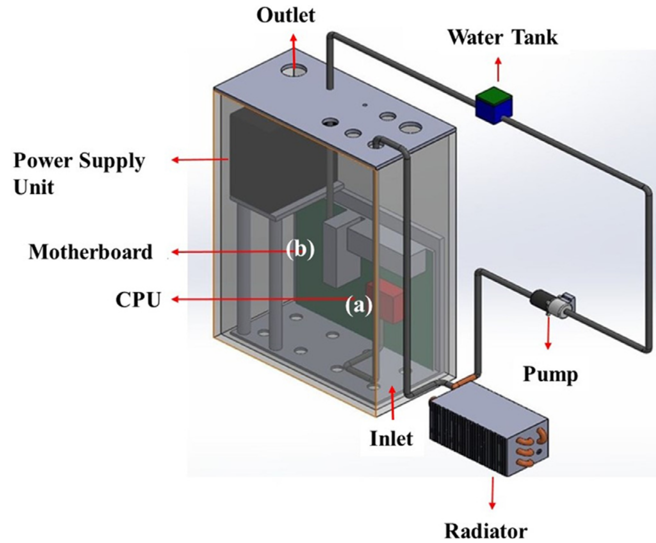

Fig. 2. Schematic drawing of the immersion cooling system for a CPU of a PC. The entire system is immersed in the coolant.

where Q is the heating source, C_p is the heat capacity, T is the temperature, q is the heat flux, and k is the heat transfer coefficient. The dimensions of the designed model is 320 mm in length with a height of 240 mm and a width of 180 mm.

2.2. Model validation

In order to verify the simulation model, an experimental structure is shown in Fig. 2. The structure includes a CPU (I9–9900 K, Intel, USA) on a mainboard (ROG MAXMIMUS X HERO, Asus, Taiwan), and a power supply unit (PSU) (CMPSU-1000HX/PHP, Corsair, USA). The I9–9900 K 8 core, 16 thread CPU with a 3.60 GHz process base frequency, and 5.00 GHz maximum turbo frequency, as well as a 95 W thermal design power (TDP), is the main source of heat within the PC. The mainboard is equipped with 4 × DIMM (maximum 64 GB), DDR4 SDRAM memory, two Aura 4-pin RGB-strip headers, intuitive Sonic Studio III and Sonic Radar III, and LGA 1151 processor socket. The PSU is equipped with a cooling fan and provides the output power of 1000 W under the operating conditions of between 100 and 240 V, 6.5–13 A, 47–63 Hz and a temperature of 50 °C.

The entire structure with the mainboard and its peripherals are immersed in the engineered fluid coolant (3 M Novec 7100, USA). The properties of engineered fluid are: coolant itself has a boiling point of 61 °C, a vaporization heat of 112 kJ/kg, a specific heat of 1.183 kJ/kg-K and a thermal conductivity of 0.069 W/m-K. After removing the heat from the CPU with the heat sink, the liquid coolant flowed out from the outlet at the top left of the PC box, passed through a water tank, pump and a radiator for cooling where it. Finally, it flowed back to the PC box through the inlet at bottom side and continued the next circulation cycle. During the experiment, the consumed power of the PC was measured by a power meter (SPG-26MS, Panasonic, Japan) within an accuracy of 0.5% when measuring power consumption, and an accuracy within 0.2% for voltage and 0.3% for current, respectively. The temperatures of liquid coolant contacting with the CPU with heat sink and under the outlet location were measured by a digital thermometer (DE-3004 Type-k, DER EE, Taiwan). The resulting temperatures were within an accuracy of ±(0.3%+1 °C) under the range of 0~1000 °C. For verifying the experimental and simulation results, the CPU with heat sink made by Al was run at a power of 50, 60, and 70 W sequentially while the circulated liquid coolant flowed

Table 1
Testing results of the four mesh types.

Mesh types	Coarser	Coarse	Normal	Fine
Mesh points	26,458	44,782	80,282	355,967
Average quality of elements	0.5518	0.5872	0.6128	0.6599

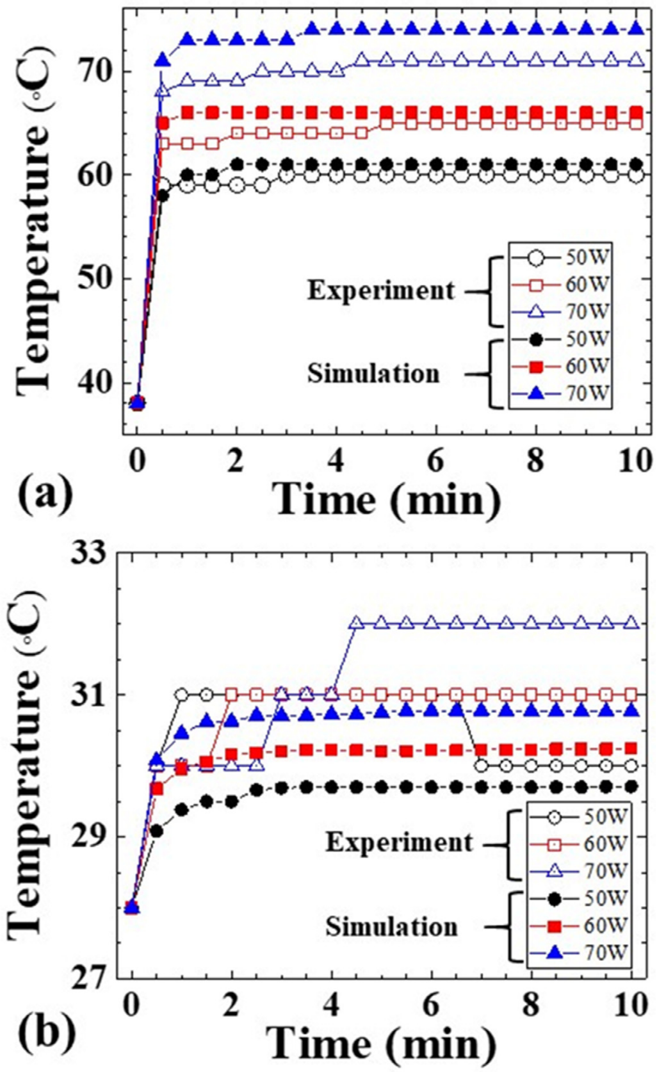

Fig. 3. Temperatures measured at front of heat sink by using the time interval of free and exact levels of 0.1, 0.2, 0.4 and 0.8 second.

at a rate of 0.8 m/s for 10 min. The measured temperatures of liquid coolant in front of CPU with heat sink and under the outlet location, as shown in Fig. 2(a) and (b), compared with the simulation results. After analysis, the heat radiation effects on the models, which the CPU with Al/Cu heat sink was run at a power of 60 W and circulated by the liquid coolant at flow rates of 0.4, 0.8 and 1.2 m/s, would be discussed further.

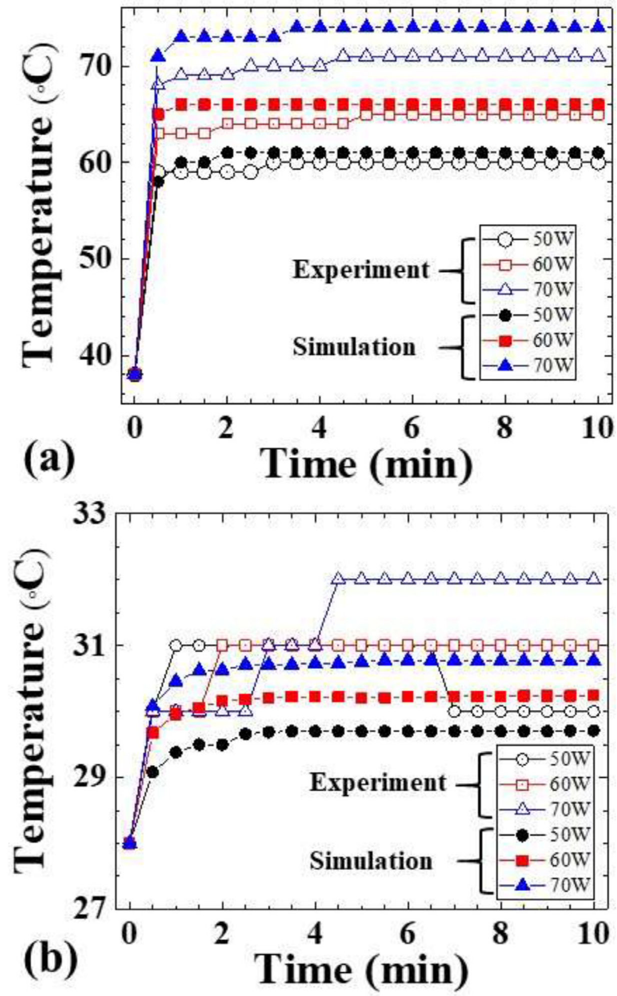

Fig. 4. Temperature comparisons between experimental and simulated results measured (a) in front of the CPU with heat sink and (b) below the outlet, as shown in Fig. 1. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

3. Results

To enhance the accuracy and resolution of the simulated results and reduce the time necessary to make calculations and lighten the demands on the PC made by running simulations, proper mesh sizes and time intervals are very important. In this study four mesh sizes of Coarser, Coarse, normal and fine, were utilized to compare the mesh points of the model and the time it takes for the PC to make calculations. The testing results of these four mesh types are listed in Table 1. The larger amount of mesh points produces better simulation results, but increases the period for PC to make calculations. The range of the average quality of elements is between 0 and 1, and larger values are better.

In Table 1, the normal mesh type illustrated has 80,282 mesh points and an average quality of 0.6128. Among these four mesh

Table 2
Mean and standard deviation of temperatures at five points around heat sink.

	Front	Top	Bottom	Right	Left
Mean temperature (°C)	300.09855	300.18024	300.02221	300.16738	300.15226
Standard deviation (10^{-4} °C)	7.83454	2.90511	0.849125	0.542748	302.2

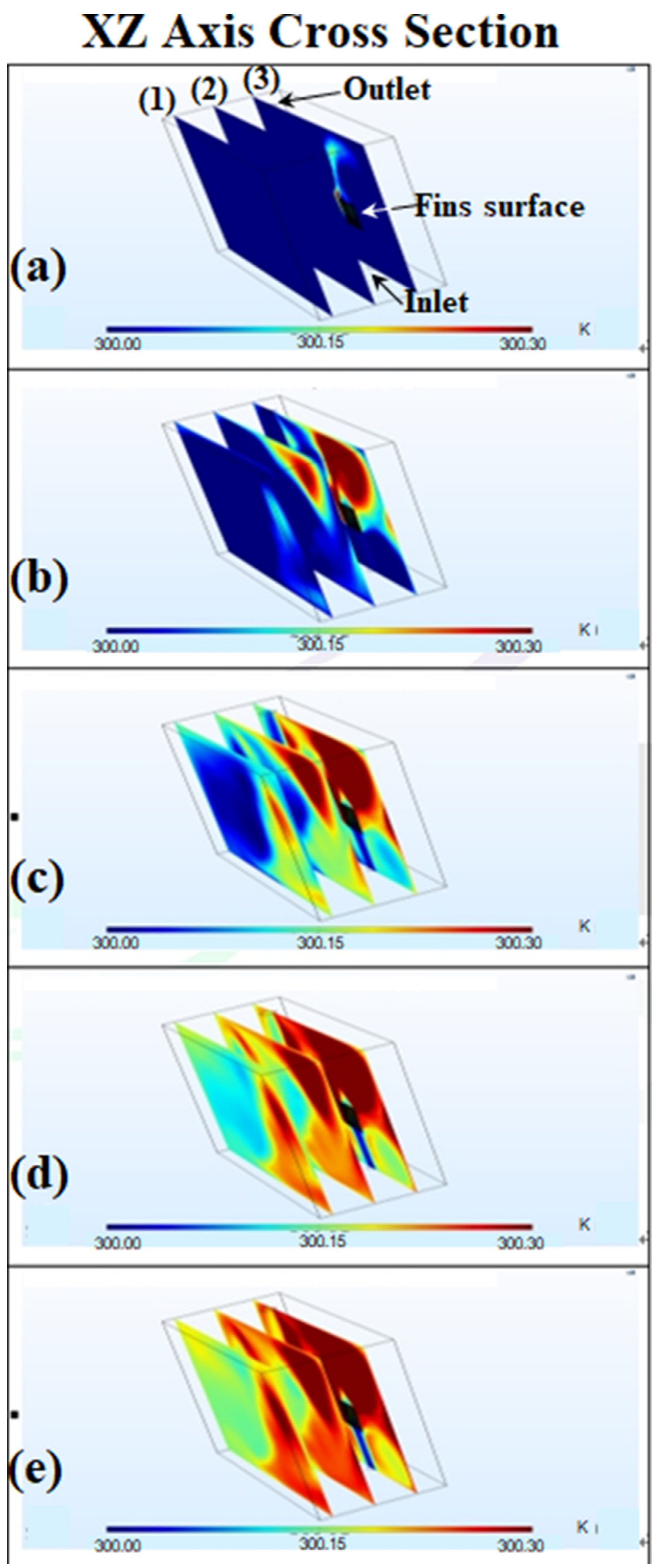

Fig. 5. A cross-sectional view showing the simulation results in the temperature variations measured in the immersion cooling system when viewed from the XZ axis shown in (a) ~ (e) from 2 to 10 min mark. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

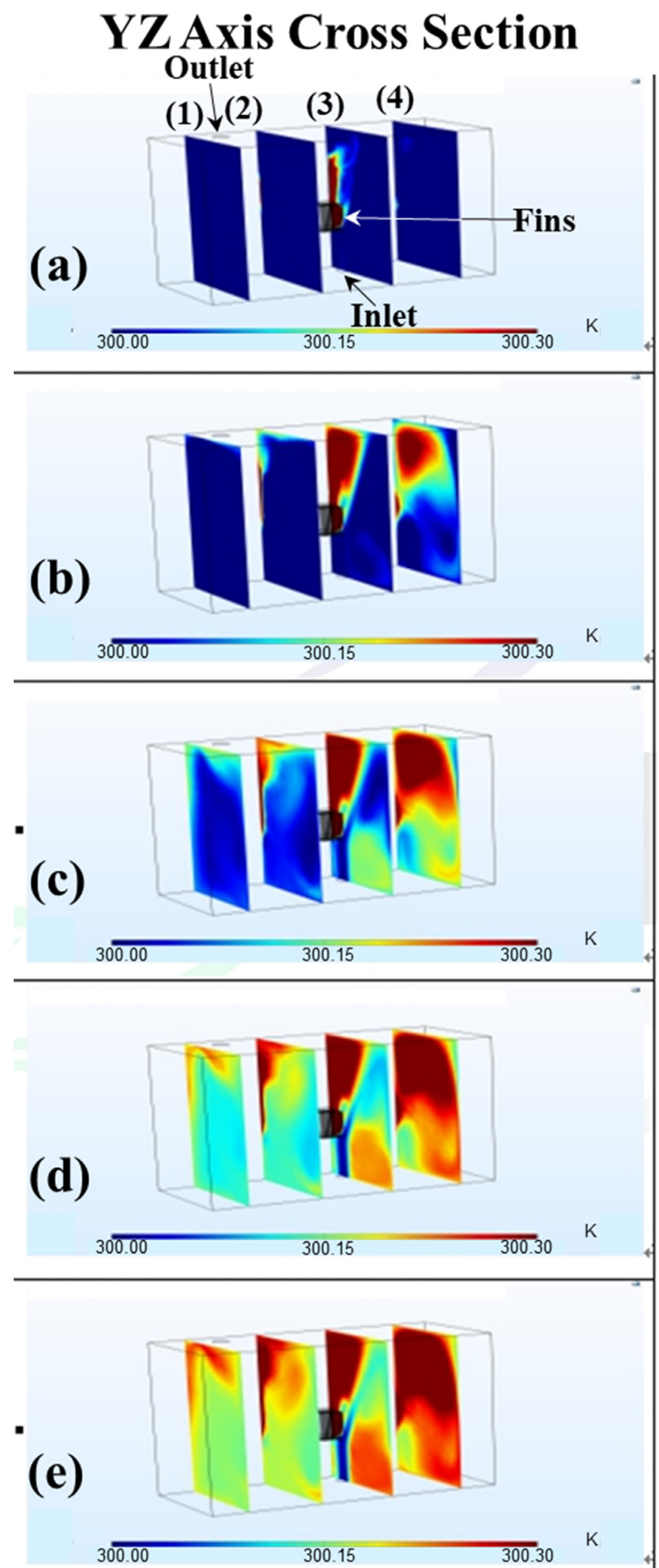

Fig. 6. A cross-sectional view of the simulation results in the temperature variations measured in the immersion cooling system when viewed from the YZ axis shown in (a) ~ (e) from 2 ~ 10 min mark. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

types, the normal type is equipped with enough of a mesh quality to make calculations in a reasonable period of time based on the capacity of the PC. After choosing the mesh type, the backward differentiation formula (BDF) of free and exact levels are used to verify the proper time intervals, so called time steps. The BDF is a linear multistep method that, for a given function and time, calculate approximately the derivative of the given function using the computed results of the previous time points for increasing the accuracy of the approximation. This method is particularly suitable for stiff differential equations and Differential Algebraic Equations (DAEs). By illustrating the temperatures measured from the front, left, right, top and bottom of the heat sink center, the simulation results utilizing the time intervals of the free and exact levels were compared. In this study, four exact time intervals of 0.1, 0.2, 0.4 and 0.8 second and free level were utilized. The distance between the heat sink center and the five points was 50 mm, as shown in Fig. 1. The temperature measured at front of heat sink by using the time interval of free and exact levels of 0.1, 0.2, 0.4 and 0.8 second, as shown in Fig. 3, were typified in symbols of square, circle, top-triangle, bottom-triangle and diamond, respectively. In Fig. 3, these temperatures simulated by using these time intervals were very consistent, and their mean value and standard deviation at the 10 min mark were 300.09885 and 7.83454×10^{-4}°C, respectively. The mean and standard deviation of temperature by using these five time intervals at these five points were listed in Table 2. In Table 2, all standard deviations of these temperatures measured at five points were less than 0.01 °C, except that at the left side of heat sink, indicating that the temperatures simulated by using these five time intervals were in consistence. Therefore, the time step at free level would be utilized for simulation in this study to reduce the calculation period.

In order to understand if the simulated results close to the experimental ones and if the model settings were proper or not, the temperatures in front of the CPU with heat sink and under the outlet location were measured, as shown in Fig. 4(a) and (b), respectively. The flow rate of the liquid coolant was 0.8 m/s, and the material of heat sink was Al. The experimental and simulated results are shown as hollow and filled symbols, respectively. The varying power levels of 50, 60, and 70 W which were run through the CPU are represented as circles, squares and triangles, respectively. In Fig. 4(a), the temperature measured in front of the CPU with heat sink rose from 38 to 59, 63 and 68°C within the 30 second mark while the CPU was running at 50, 60 and 70 W, respectively. These temperatures reached their peak values of 60, 65 and 71 °C after 4 min and 45 second mark. The simulated results also expressed similar tendencies. The temperature differences between the simulated and experimental results at end of the experiment increased along with the power that was run through the CPU. They were 1, 1.2 and 3 °C (or 1.7, 1.5 and 4%) for the CPU running at 50, 60 and 70 W, respectively.

In Fig. 4(b), the temperatures measured under the outlet location rose from 28 to 30 °C within the 30 second mark, and reached their final values of 30, 31 and 32 °C with the CPU running at 50, 60, and 70 W respectively. The temperature variations at this location were smaller than those measured in front of CPU with heat sink, due to the circulation of the liquid coolant. This will be discussed further in the following sections. The simulated results expressed similar tendencies. The temperature differences between the simulated and experimental results at end of experiment increased as the power to the CPU was increased. The temperature differences measured were 0.29, 0.75 and 1.23 °C (or 1, 2.5 and 3.9%) when the CPU was running at 50, 60, and 70 W, respectively. It is noted that the simulated model neglected the heat produced from the mainboard and the several capacitors that are on the memory unit. This is the reason why the simulated temperatures under the outlet location were less than the experimental

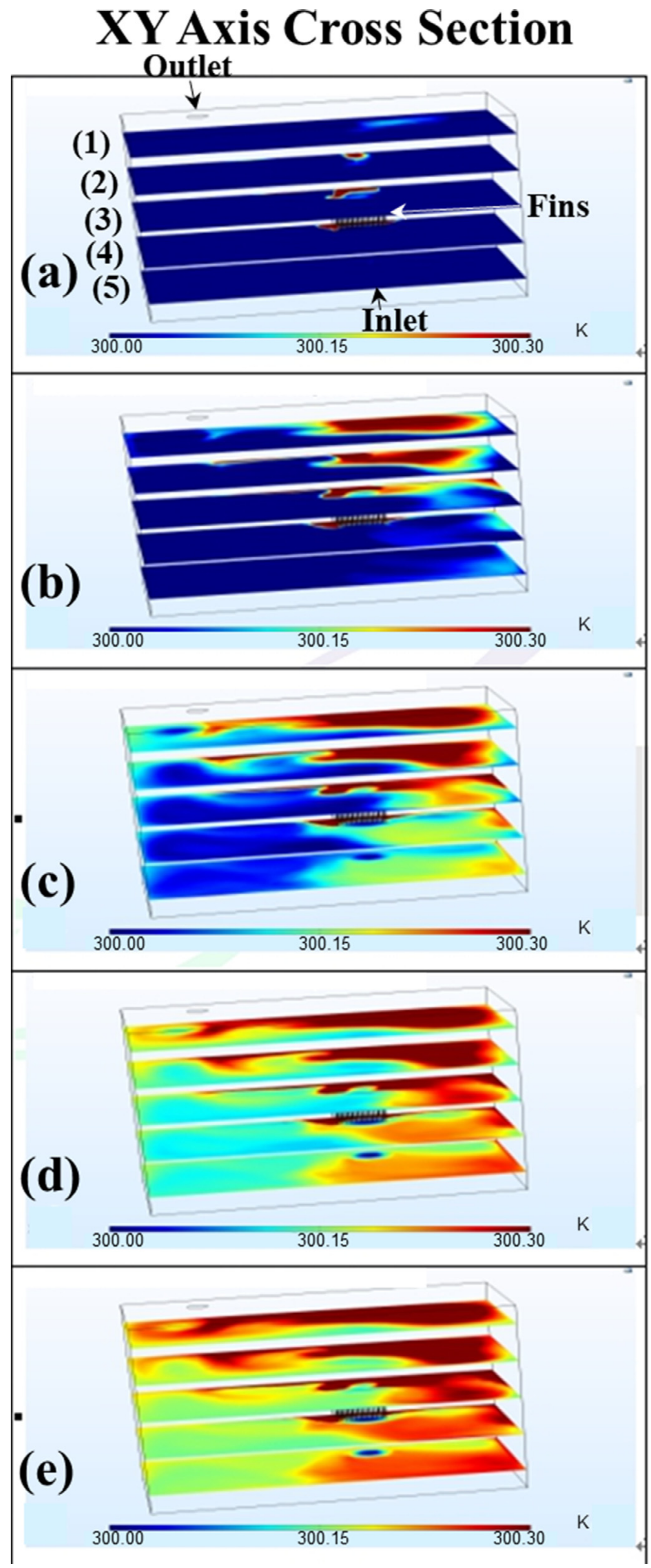

Fig. 7. A cross-sectional view of the simulation results of the temperature variations measured in the immersion cooling system when viewed from the XY axis shown in (a) ~ (e) from 2 ~ 10 min mark. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

ones. Even though, the differences between the simulated and experimental temperatures were still within the acceptable ranges. Therefore, this model can be utilized to simulate the heat radiation

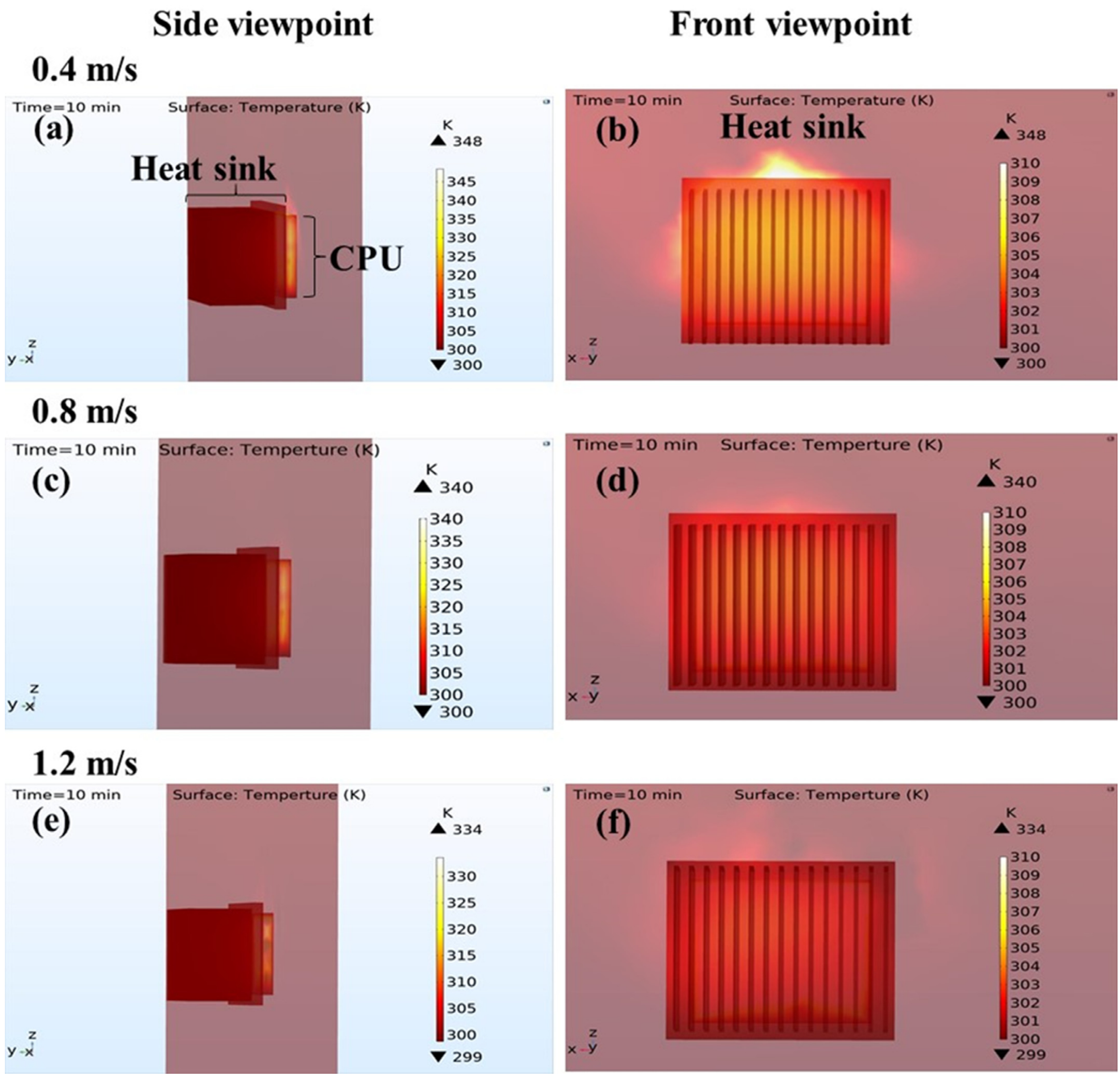

Fig. 8. Front and side-views of the simulation results in temperature variations within the immersion cooling system at the 10 min mark at flow rates of (a) ~ (b) 0.4 m/s, (c) ~ (d) 0.8 m/s, and (e) ~ (f) 1.2 m/s, respectively.

effects of a CPU with various materials of heat sink and circulation speeds of the liquid coolant.

4. Discussions

4.1. Heat distribution from various angles

Based on the designed model, showing the distribution of the heat radiation around the CPU from various angles can guide the readers to better understand how the liquid coolant and heat sink remove heat. In this section, an example of CPU with heat sink running at 60 W and cooled at a flow rate of 0.4 m/s was observed. The various angles of heat distribution are shown in cross-sections of the XZ, YZ, and XY axis of the designed model in Figs. 5, 6, and 7 respectively. It is noted that the temperature mark was from 300.00 to 300.30 K for these three figures. Fig. 5(a) ~ (e) illustrates the flow state and temperature variations of the liquid coolant during a process time from the 2 to 10 min marks, respectively. There were three cross-sections on the XZ axis, as indicated in Fig. 5(a). The 3rd cross-section shows a location that is past the inlet, outlet and heat sink surfaces. The liquid coolant flowed in the model through the inlet, passed the heat sink and increased the temperature of the area located at the top of the heat sink. The heat also expanded into the 2nd cross-section. In Fig. 5(b), the heat expanded further into the surrounding area, and even to the left side of CPU and into the 1st cross-section. In Fig. 5(c) ~ (e), the heat distribution reached a stable state. The temperature

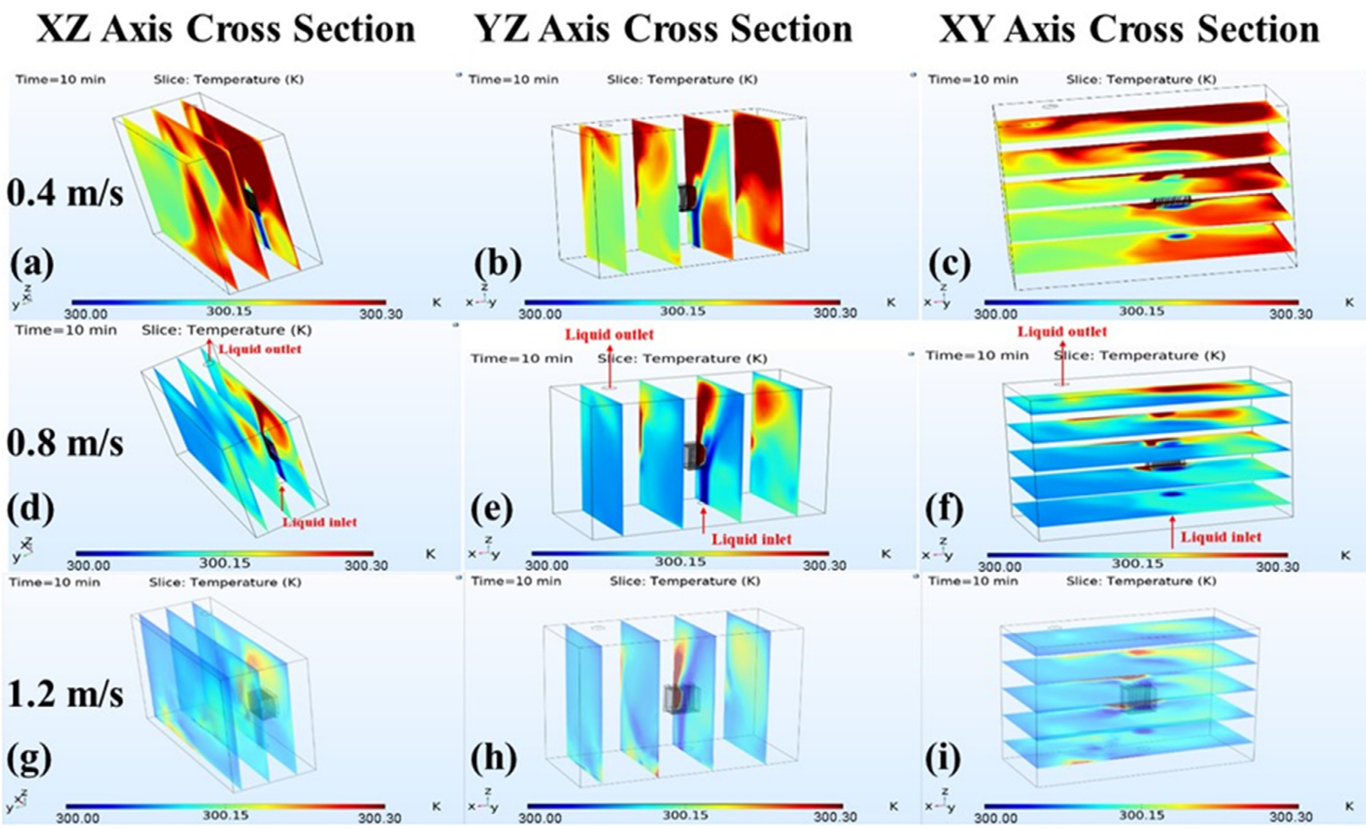

Fig. 9. Cross-sectional views of the simulation results showing the temperature variations in the immersion cooling system operating at the 10 min mark with a coolant flow rates of (a) ~ (c) 0.4 m/s, (d) ~ (f) 0.8 m/s, and (g) ~ (i) 1.2 m/s, respectively.

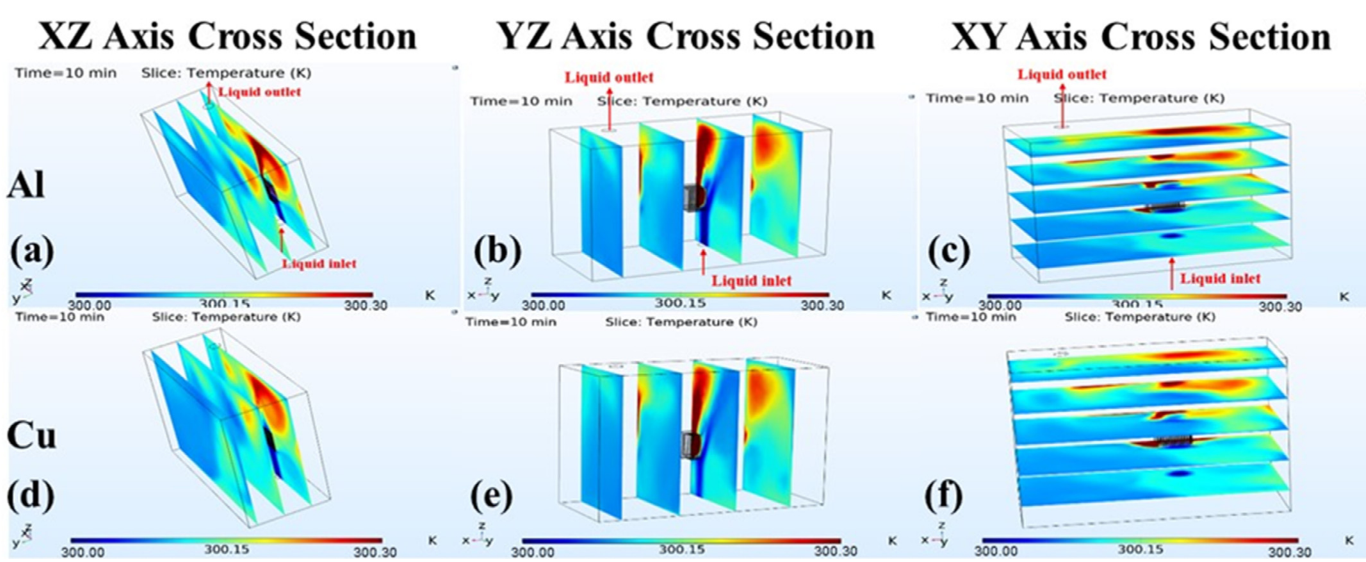

Fig. 10. Cross-sectional views of the simulation results of temperature variation of immersion cooling system operated at the 10 min mark with a coolant flow rates of 0.8 m/s. Heat sinks of CPU are made by (a) ~ (c) Al and (d) ~ (f) Cu, respectively.

measured at the top of CPU was the highest. The temperature on the right side of the CPU was higher than that of the left side. The liquid coolant that flowed from the inlet had the lowest temperature.

Fig. 6(a) ~ (e) illustrates the flow state and temperature variations of the liquid coolant on the YZ axis during a process time from the 2 to 10 min mark, respectively. There are four cross-sections on the YZ axis from left to right. The 3rd cross-section shows a location that is past the heat sink and the inlet. The outlet location is located between 1st and 2nd cross-sections. In Fig. 6(a), the 3rd cross-section shows the liquid coolant flowing from the inlet, passing the heat sink and removing the heat to the top of the heat sink in a bar shape. The removed heat also expanded into the 4th cross-section. In Fig. 6(b), the removed heat expanded further to the 1st and 2nd cross-sections. In Fig. 6(c) ~ (e), the removed heat reached a stable state. The top of the 3rd cross-section had the highest temperature. The liquid coolant flowing from the inlet had the lowest temperature. The temperature in the 4th cross-section was higher than those in the 1st and 2nd cross-sections.

Fig. 7(a) ~ (e) shows the flow state and temperature distribution of the liquid coolant on the XY axis during a process time from the 2 to 10 min mark, respectively. There are five cross-sections on the XY axis, and the 4th cross-section is located past the heat sink. In Fig. 7(a), the first 4 cross-sections show that the liquid coolant passed through the heat sink and removed the heat towards the right side of 1st cross-section. In Fig. 7(b), the removed heat expanded towards the direction of the left side of 1st cross-section and towards the direction of the right side of the 5th cross-section. In Fig. 7(c) ~ (e), the removed heat reached a stable state. The highest temperature appeared at the top and right side of the heat sink. The area from the heat sink to the outlet also had a higher temperature. The temperature at the right side of the heat sink was higher than that of the left side. The liquid coolant that flowed from the inlet to the heat sink had the lowest temperature.

4.2. The heat distribution under various flow rates of the liquid coolant

After attaining a clear picture of the heat distribution from the XZ, YZ, and YX axis cross-sectional views of the designed model, it would be quite valuable to also investigate the effect the flow rate of the liquid coolant has on the heat radiation of the CPU and heat dispersion throughout the liquid coolant. Fig. 8 shows a front-side view of the temperature variations around the CPU with heat sink during different flow rates of the liquid coolant of 0.4, 0.8 and 1.2 m/s at the 10 min mark of the process time. The material of heat sink was Al, and the CPU was running at 60 W. Fig. 8(a), (c) and (e) illustrates a side-view of the temperature variations around the CPU at flow rates of 0.4, 0.8 and 1.2 m/s, respectively. The highest temperatures of CPU at the 3 different flow rates were 348, 340 and 334 K, respectively. The faster flowing speed of liquid coolant would remove more heat, and cause the lower temperature of CPU. The temperature measured on the heat sink surface was ranged from 300 to 305 K. Fig. 8(b), (d) and (f) illustrates a front-view of the temperature variations around the heat sink at flow rates of 0.4, 0.8 and 1.2 m/s, respectively. The area at the bottom of the heat sink surface showed the lowest temperatures, due to being in direct contact with the liquid coolant. The heat of the heat sink that originated from the CPU was removed by the liquid coolant in a flow direction from the bottom towards the top. The slower flow rate of the liquid coolant caused the more concentrated heat around the heat sink. The temperatures measured at the center of heat sink surface under the 3 different flow rates were 307, 305 and 303 K, respectively.

Fig. 9 shows the heat distribution around the CPU with heat sink from XZ, YZ, and YX axis cross-sectional views at the 10 min mark at flow rates of 0.4, 0.8 and 1.2 m/s. In the XZ axis cross-section in Fig. 9(a), (d) and (g), for the heat distribution at a flow rate of 0.4 m/s, the highest temperature that was measured was located at the location above the heat sink. The heat dispersion between left and right sides was unbalanced, and the heat movement was at the slowest speed. At a flow rate of 0.8 m/s, the highest temperatures were still measured at the top of heat sink, but the unbalanced heat distribution had improved. The overall reduced temperature values present in all three of the cross-sections indicate improved heat dispersion. At a flow rate of 1.2 m/s, the highest temperatures were still measured at the top of heat sink, but the most improvements in terms of reducing imbalances in heat dispersion and the speed at which heat is displaced can be seen at this flow rate.

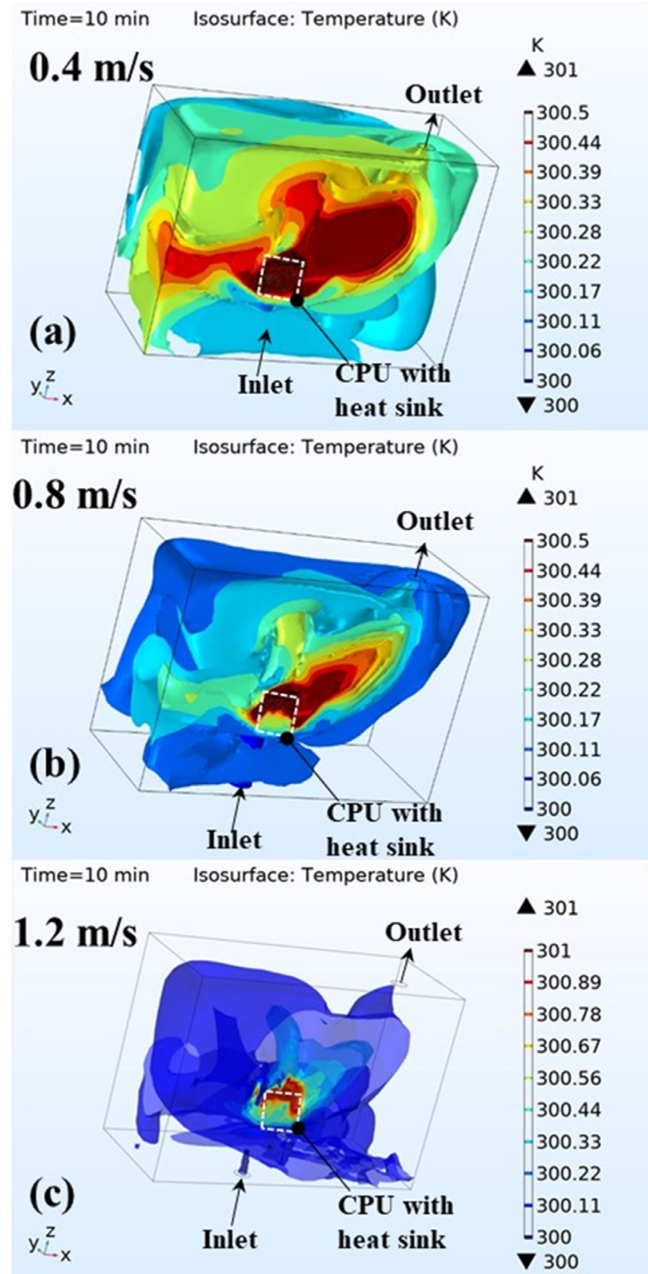

Fig. 11. 3D isothermal surface model of the immersion cooling system observed at the10 min mark at liquid coolant flow rates of (a) 0.4 m/s, (b) 0.8 m/s, and (c) 1.2 m/s, respectively.

When analyzing the heat distribution of the YZ axis cross-sections shown in Fig. 9(b), (e) and (h), at a flow rate of 0.4 m/s, the highest temperatures were measured at the top and the front of the heat sink. Additionally, there was a clear evidence of heat dispersion imbalances surrounding the heat sink. There were also areas congestion that were shown by the turbulence in the flow at the top left and bottom right of the model. The highest temperatures were measured at the top and front of the heat sink at flow rates of 0.8 and 1.2 m/s. However, as the flow rate was increased, heat dispersion imbalances, average overall temperatures, and congestion in the flow was reduced.

When observing the heat distribution of the XY axis cross-sections shown in Fig. 9(c), (f) and (i), the heat dispersion imbalances surrounding the heat sink were quite obvious with a liquid coolant flow rate of 0.4 m/s. However, as the flow rate was in-

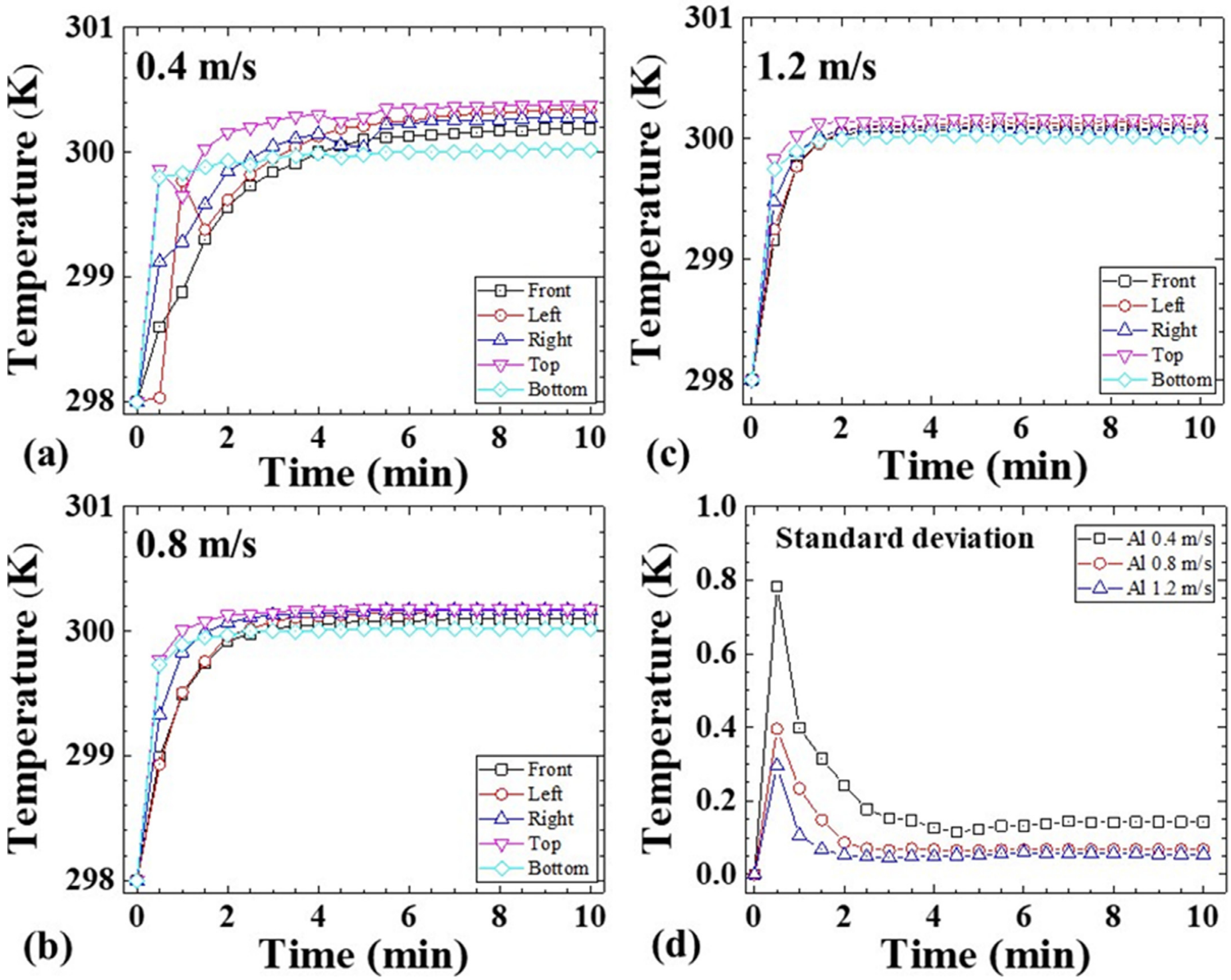

Fig. 12. Simulation results of the temperature variations in the immersion cooling system with coolant flow rates at (a) 0.4 m/s, (b) 0.8 m/s, and (c) 1.2 m/s, respectively.

creased, the heat dispersion imbalances were decreased and a noticeable improvement in the overall flow of the coolant within the system appeared.

4.3. Heat distribution with heat sink made by al and cu

Having observed and compared the effects of different flow rates of the liquid coolant, this study is also interested in observing the effects that heat radiation may have on heat sink made of different materials. Fig. 10 shows the simulated heat distribution surrounding a CPU with heat sink made of Al and Cu. The observations can be seen in the three different XZ, YZ and XY axis cross-sections of the simulation model at the 10 min mark of the process time with a coolant flow rate of 0.8 m/s. The heat distribution of the Al heat sink shown in Fig. 10(a) ~ (c) has already been described in the above sections and comparatively, shown in Fig. 10(d) ~ (f), a similar heat distribution pattern was also observed using Cu heat sink, except that the highest temperature readings from the Cu heat sink were less than those of the Al heat sink. This is due to the fact that the heat transfer coefficient of Al is 237 W/m-K, and that of Cu is 401 W/m-K. Therefore, the Al heat sink has the capacity to dissipate more heat to the liquid coolant than the Cu heat sink.

4.4. Three-dimension (3D) isothermal surface model

Using a 3D isothermal surface model under various flow rates of liquid coolant can provide useful data regarding the flow path, heat distribution and temperature gradient of liquid coolant. Fig. 11(a) ~ (c) shows 3D isothermal surface models of CPU with Al heat sink, as indicated by the dotted line, at different flow speeds of 0.4 ~ 1.2 m/s, respectively. In this 3D model, the liquid coolant flowed into the PC box through the inlet, came into contact with the CPU and heat sink, and expanded into the surrounding area. Along with the flowing path, part of the liquid coolant with the removed heat flowed out through the outlet. The rest of the liquid coolant remained at the bottom left side of the CPU. When comparing Fig. 11(a) ~ (c), it can be observed through the yellow and red areas that the slower flow rate caused the liquid coolant to stagnate, causing the higher average temperatures, larger temperature gradient and temperature dispersion imbalances as described in the above sections.

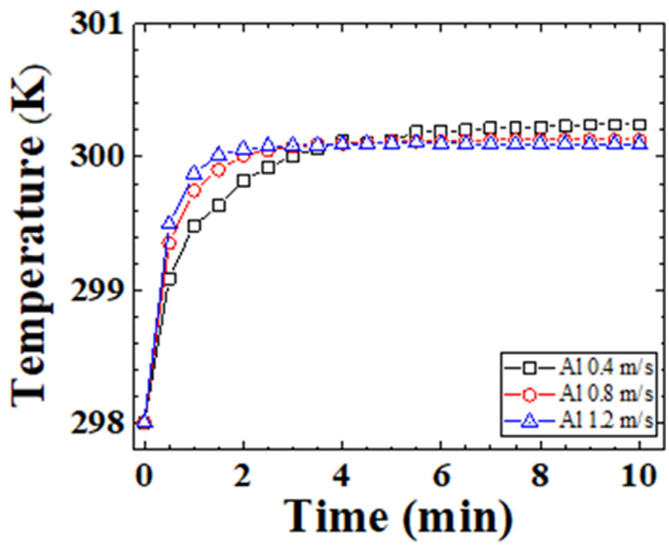

Fig. 13. Comparison of average temperature of immersion cooling system with the flowing speed at 0.4, 0.8 and 1.2 m/s.

4.5. Temperatures measured at five points around the cpu and heat sink

After providing a detailed picture of the heat distribution through analysis of cross-sections of the simulation models and isothermal models, this study was also able to analyze the various temperatures that are given off through the heat radiation process from five measured points around the CPU. Five points at the front, right, left, top and bottom of the CPU were chosen and the temperatures of these points were measured continuously as liquid coolant at flow rates of 0.4, 0.8 and 1.2 m/s were cycled throughout the system. The measured temperatures at various flow rates are presented in Fig. 12(a), (b) and (c). The temperatures measured at the front, left, right, top and bottom of the CPU with Al heat sink were typified in symbols of square, circle, up-triangle, down-triangle and diamond, respectively. In Fig. 12(a) at a flow rate of 0.4 m/s, the temperatures measured at five points increased from 298 to 300 K monotonically within a process time of 5.5 min mark. At the 10 min mark, the highest and lowest temperatures were 300.02 and 300.38 K, which were measured at top and bottom of heat sink, respectively. The temperature difference was 0.36 K. This evidenced the temperature dispersion imbalances surrounding the heat sink, as mentioned in the previous sections. In Fig. 12(b) and (c) at flow rates of 0.8 and 1.2 m/s, the temperatures measured at the five points increased from 298 to 300 K monotonically within the process time of 3.0 and 2.0 min mark, respectively. At the 10 min mark, the highest and lowest temperatures were 300.18, 300.02 and 300.17, 300.02 K at these two flow rates. It is obvious that the heat dispersion imbalances and average overall temperature are reduced as the flow rate is increased.

The standard deviations of the measured temperatures at these five points at the different flow rates of liquid coolant are shown in Fig. 12(d). The standard deviations of the measured temperatures at flow rates of 0.4, 0.8 and 1.2 m/s were typified in symbols of square, circle and up-triangle, respectively. At the 30 second mark, the standard deviations of measured temperatures were 0.78, 0.39 and 0.29 K at flow rates of 0.4, 0.8 and 1.2 m/s, respectively. At the 2 min mark, the standard deviations of the temperature variations were 0.24, 0.09 and 0.05 K at the three different flow rates. The settling times, which was the time when the standard deviation reached the final value, for the three flow rates were at the 6.5, 4.0 and 1.5 min-mark, respectively. The final standard deviations of the temperature variations was 0.14, 0.07 and 0.05 K for each flow rate. The temperature variation and settling time were reduced when the flow rate of liquid coolant was increased.

The average temperatures measured at the five points when subjected to different flow rates of liquid coolant are shown in Fig. 13. The average temperatures measured at flow rates of 0.4, 0.8 and 1.2 m/s were typified in symbols of square, circle and up-triangle, respectively. At the 1 min mark, the average temperatures were 299.87, 299.75 and 299.48 K at flow rates of 1.2, 0.8 and 0.4 m/s, respectively. The temperatures measured at settling times at flow rates of 1.2, 0.8 and 0.4 m/s were 300.01, 300.10 and 300.20 K, respectively. The results indicate that, with the increasing flow rate of liquid coolant, the settling time and the average overall temperature of the design model are reduced. Therefore, the heat balance situation of coolant during the experiment is an indicator of proper flow rate of circulation coolant to guarantee the safe operation of CPU.

5. Conclusions

In this study, an innovative cooling structure and procedure of a single-phase immersion cooling system with heat sink and forced circulation is presented. The straight finned heat sink was attached on to a CPU's surface and then the entire mainboard was submerged into the 3 M Novec 7100 engineered fluid for forced circulation cooling. The computer simulation software was used to build a simulation model to verify results of the experiment. Three circulating flow rates of coolant and two materials of Al and Cu for the heat sink were chosen to explore the cooling effect of the single-phase immersion cooling system. The purpose of the experiment was to determine what would be the most suitable material of the heat sink and circulating conditions of coolant by simulating coolant flowing through this single-phase immersion cooling system and analyzing the temperature variation at various flow rates. The heat distribution of the designed model at various flow rates of liquid coolant and materials of heat sink was observed through the cross-sections on the XZ, YZ and XY axis and 3D isothermal surface model. The results of the simulations show that there was the unbalanced heat distribution around CPU, and the faster flowing speed of liquid coolant was able to remove more heat, then cause the lower temperature of CPU and more balanced heat distribution around CPU. However, the slower flow rates caused the liquid coolant to stagnate in the system, which in turn, caused higher heat distribution imbalances. The highest temperatures were measured at the top of the heat sink, and the temperature value of coolant on the right side of heat sink was higher than that of the left side. For the temperature variation of coolant around CPU during cooling process, the settling times and temperatures under the flowing speeds of 1.2, 0.8 and 0.4 m/s were 1.5, 4.0 and 6.5 min and 300.01, 300.10 and 300.20 K, respectively. The final standard deviation of temperature variations were 0.05, 0.07 and 0.14 K for these three flowing speeds. The temperature variation and settling time were reduced when the flow rate of liquid coolant was increased. Therefore, the heat balance situation of coolant during the experiment is an indicator of proper flow rate of circulation coolant to guarantee the safe operation of CPU. It is noted that the material of heat sink does not affect the results significantly. These outcomes are able to provide the system designer with useful information to improve power densification and guarantee the safe operation of the related computer, server and communication systems.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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